

HABITATS FOR LIFE

Earth's uniquely varied habitats enable it to support an abundantly rich diversity of animal and plant life – from the extreme depths of the ocean floors to the highest mountains, and from arid deserts and grasslands to the warm, wet tropics.

Each life form has its preferred habitat – the one to which it has adapted over thousands or even millions of years. However, Earth's varied environments allow many different kinds of animals and plants to live in the same habitat – a phenomenon known as biodiversity. Over geological time, life has had to adapt to environmental change, especially rapid climate change, with both extinctions and the evolution of life forms to colonize new habitats. The presence of these pioneering organisms in turn produced changes in the environments they colonized, by forming soils for instance, and these modifications encouraged further colonization by new life forms.

Differences in habitats are produced by many factors, ranging from an area's height above sea level, its distance from the equator, and its topography (or physical shape). Some areas of the globe are biodiversity "hotspots", rich in animal and plant life – most notably tropical reefs and rainforests – while other areas with more extreme conditions support only a few, though often richly populated, species.

SCIENCE

LEVELS OF LIFE

Organisms rarely exist entirely on their own, even in the remotest places on Earth. The interactions between wildlife create different levels – from the individual organism to the overall ecosystem which it inhabits and shares with others.



INDIVIDUAL

A single, usually independent and habitat-restricted member of a population.



POPULATION

A group of individuals of the same species that occupy the same area and interbreed.



COMMUNITY

A naturally occurring collection of plant and animal populations living within the same area.

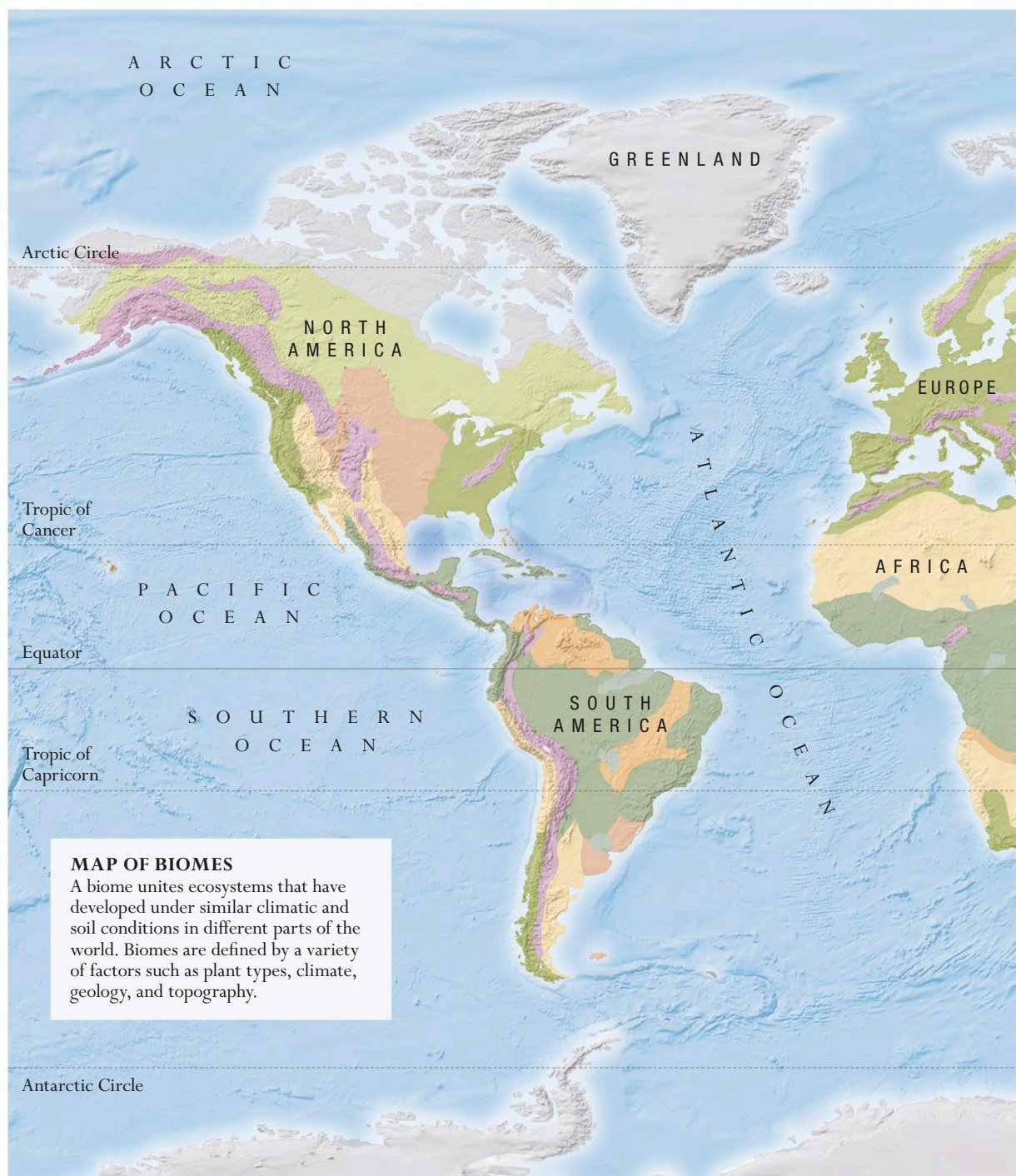


ECOSYSTEM

A biological community and its physical surroundings, which support one another.

KEY

	Polar region		Coniferous forest
	Desert		Mountains
	Grassland		Coral reef
	Tropical forest		Rivers and wetland
	Temperate forest		Oceans



MAP OF BIOMES

A biome unites ecosystems that have developed under similar climatic and soil conditions in different parts of the world. Biomes are defined by a variety of factors such as plant types, climate, geology, and topography.



GRASSLAND

The evolution of grass plants some 20 million years ago, and the colonization by grazing mammals, transformed Earth's landscapes. Temperate grasslands are generally treeless and have extremely fertile soils. Savanna grasslands, as shown here, are more like open woodland, featuring scatterings of trees and scrubs.



BISON